

## a. Goal State

$v\ w : \text{Nat} \quad x\ y : \text{BV } w$   
 $\vdash x + y = y + x$

## b. Dep. Typed Reflected Goal State

(eq (#bv 2) (#w 2)  
  (add (w 1)  
    (bv 0) (bv 1))  
  (add (w 1)  
    (bv 1) (bv 0))))

## b. Untyped Reflected Meta State

(eq  
  (add  
    (bv 0) (bv 1))  
  (add  
    (bv 1) (bv 0)))

## c. Symbolic FSM

## d. SAT Solve

## e. Add Cert to Env

def safetyCert  
def indCert

## f. Final Object Level Proof

toProp\_of\_KInductionCircuits  
  safetyCert  
  indCert  
  fsm  
  ...